

Task for Seizure Classification

Task 1: Understand the dataset given in Description.docx file. Identify the .zip folders for five classes and extract the files into 5 sets. Refer to the data extraction code provided in a Jupyter file separately.

Task 2: Organize the sets into relevant classes for binary classification. Hint: Refer to the description file for identifying the classes.

Task 3: Create a dataset consisting of two classes and the corresponding labels.

Task 4: Visualize the dataset.

Task 5: Identify the missing values in a dataset. If any, use imputation technique discussed in the class.
Note: Do not use filtering as the signals are already passed through a filter.

Task 6: Identify the number of channels. If multiple channels are there you can select 1 or many based on your choice. If single channel is there, proceed to feature extraction.

Task 7: Extract the relevant features.

Task 8: Perform feature selection either with PCA, or recursive feature elimination, or Kruskal Wallis test. Note: Remember to standardize the data first.

Task 9: Perform as train-test split.

Task 10: Initialize the classifier of your choice.

Task 11: Train and validate the classifier and plot confusion matrix and create a performance report.

Task 12: Perform cross validation and Grid search.

Task 13: Provide a comprehensive comparison of holdout, cross validation, and Grid search validation.

Task 14: Interpret the results with your peers.